

Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)



Compound angle formula for cosine

- 1 Use the compound angle formula $\cos(A + B) = \cos A \cos B - \sin A \sin B$ to find the exact value of $\cos 75^\circ$, writing $75^\circ = 45^\circ + 30^\circ$.
-

Compound angle formula for sine

- 2 Use the compound angle formula $\sin(A - B) = \sin A \cos B - \cos A \sin B$ to find the exact value of $\sin 15^\circ$, writing $15^\circ = 45^\circ - 30^\circ$.
-

Compound angle formula for tangent

- 3 The compound angle formula for tangent is $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$. Use it with $A = 45^\circ$ and $B = 30^\circ$ to find the exact value of $\tan 15^\circ$, giving your answer in simplified surd form.
-

Proving identities using the compound angle formulas

- 4 Prove that $\sin(A + B) + \sin(A - B) = 2\sin A \cos B$.
- 5 Prove that $\cos(A + B)\cos(A - B) = \cos^2 A - \sin^2 B$.
-

Solving a trigonometric equation using a compound angle formula

- 6 Solve $\sin(x + 30^\circ) = \cos x$ for $0^\circ \leq x \leq 360^\circ$.
-

Evaluating reciprocal trigonometric functions from a given ratio

- 7 Given $\cos \theta = \frac{3}{5}$ with θ in Quadrant I, find the exact values of $\sec \theta$, $\csc \theta$, and $\cot \theta$.
-

Reciprocal trigonometric functions in another quadrant

- 8 Given $\tan \theta = -2$ with θ in Quadrant II, find the exact values of $\sec \theta$, $\csc \theta$, and $\cot \theta$.

Deriving $1 + \tan^2\theta = \sec^2\theta$

- 9** Starting from the Pythagorean identity $\sin^2\theta + \cos^2\theta = 1$, divide through by $\cos^2\theta$ to derive $1 + \tan^2\theta = \sec^2\theta$.
-

Deriving $1 + \cot^2\theta = \csc^2\theta$

- 10** Starting from the Pythagorean identity $\sin^2\theta + \cos^2\theta = 1$, divide through by $\sin^2\theta$ to derive $1 + \cot^2\theta = \csc^2\theta$.
-

Simplifying an expression using reciprocal identities

- 11** Simplify $\sec\theta - \cos\theta$, expressing your answer in terms of $\sin\theta$ and $\tan\theta$.
-

Mixed application: combining reciprocal identities across quadrants

- 12** Given $\sec\theta = 13/5$ with θ in Quadrant IV, find the exact value of $\sin\theta + \cot\theta$.